

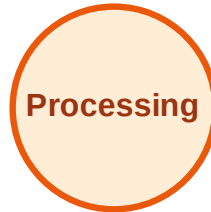
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing

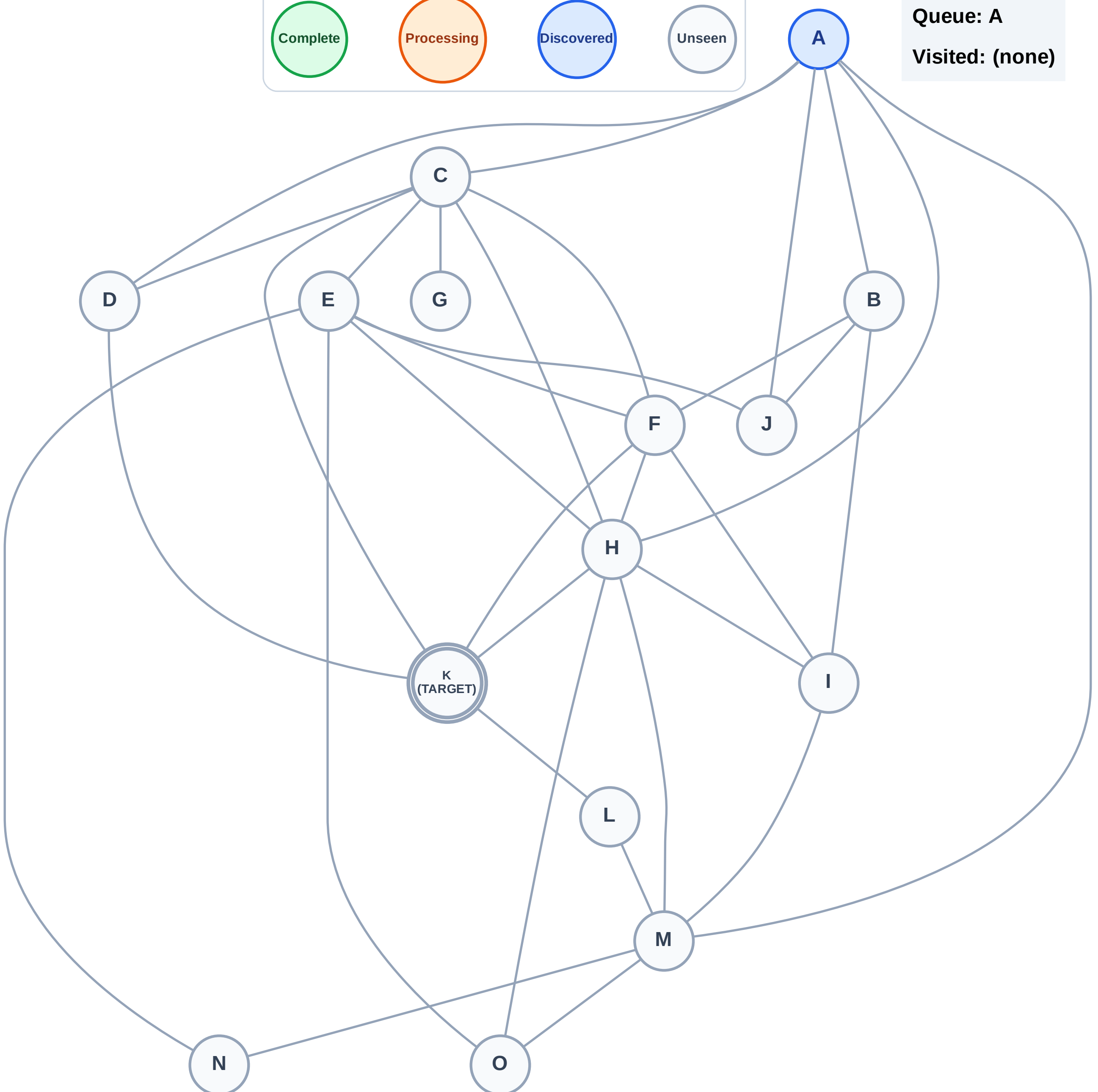


Discovered



Unseen

Queue: A
Visited: (none)



BFS Traversal

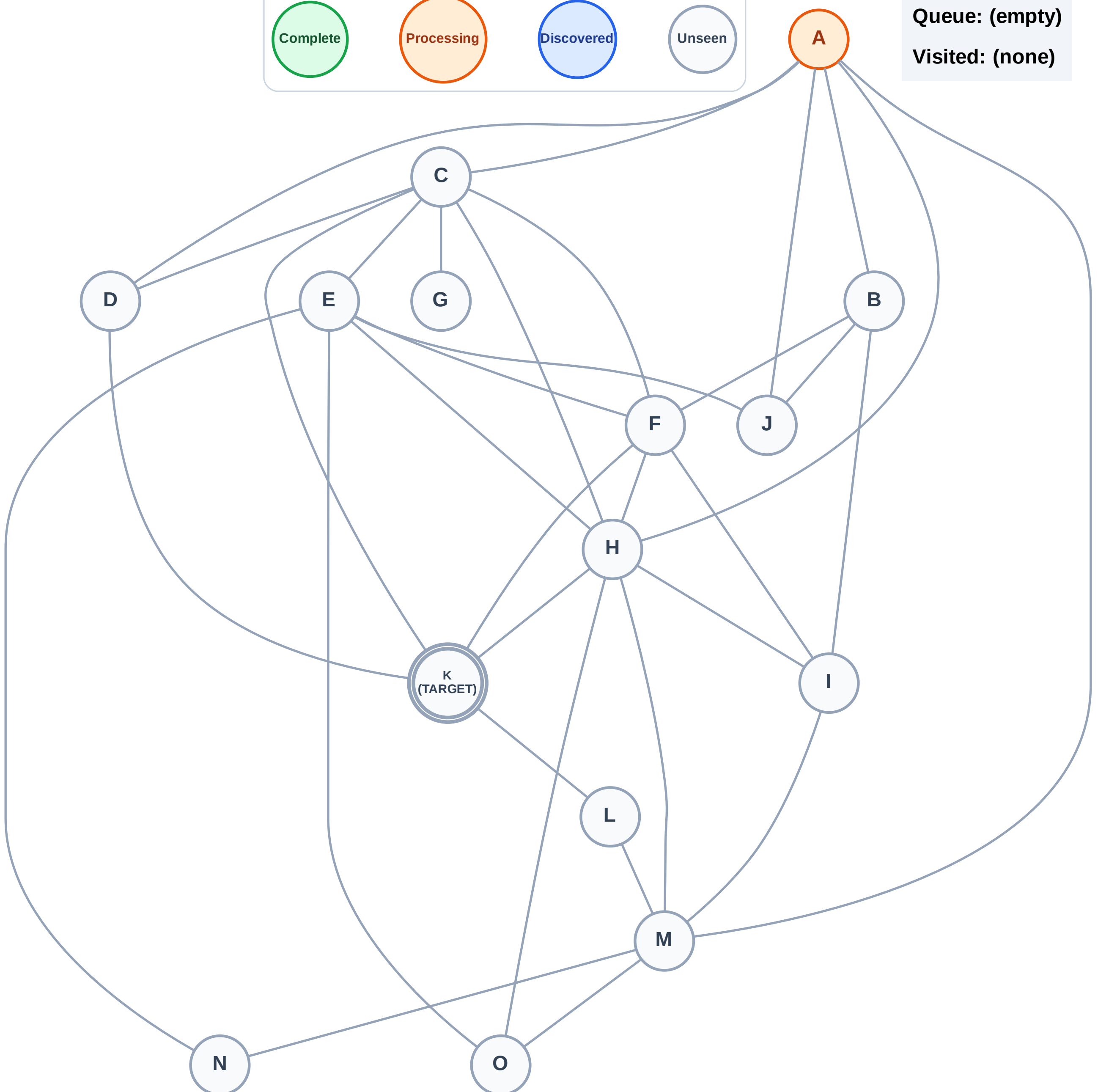
Step 2: Process node A

Legend



Queue: (empty)

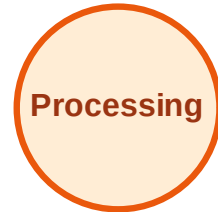
Visited: (none)



BFS Traversal

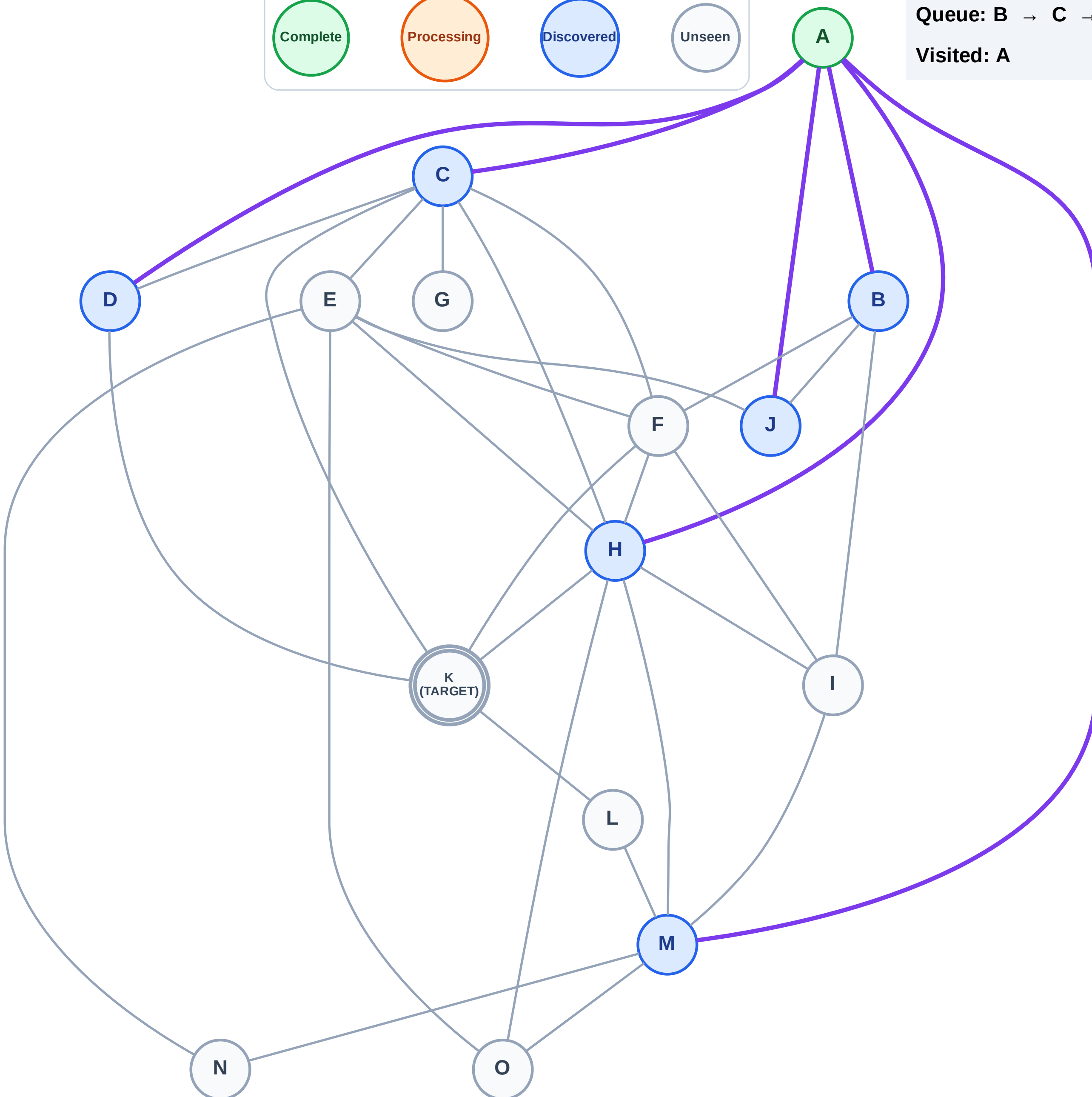
Step 3: Discover B, C, D, H, J, M from A

Legend



Queue: B → C → D → H → J → M

Visited: A



BFS Traversal

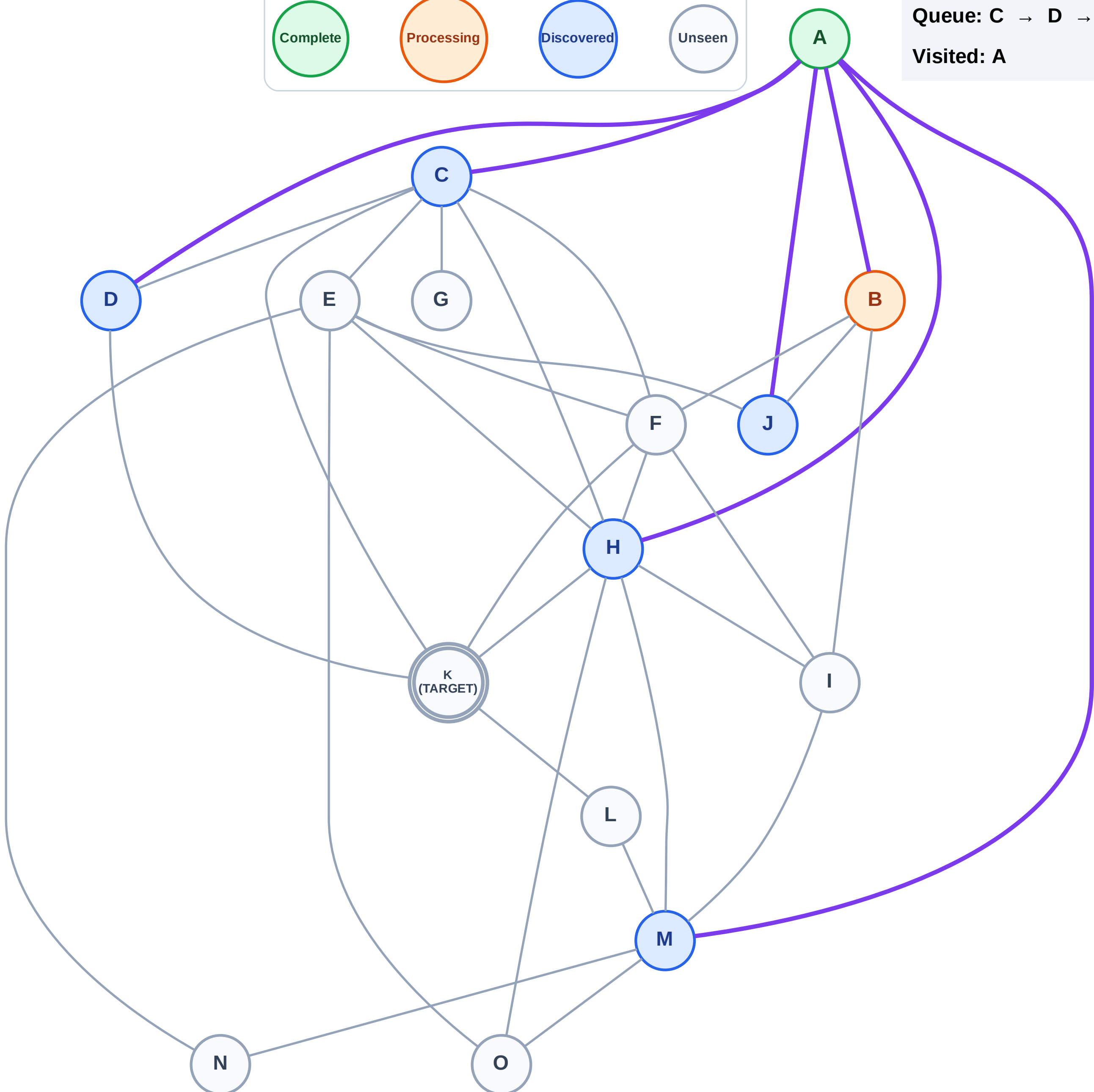
Step 4: Process node B

Legend



Queue: C → D → H → J → M

Visited: A



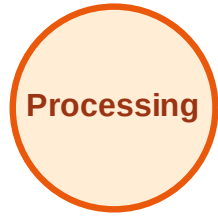
BFS Traversal

Step 5: Discover F, I from B

Legend



Complete



Processing



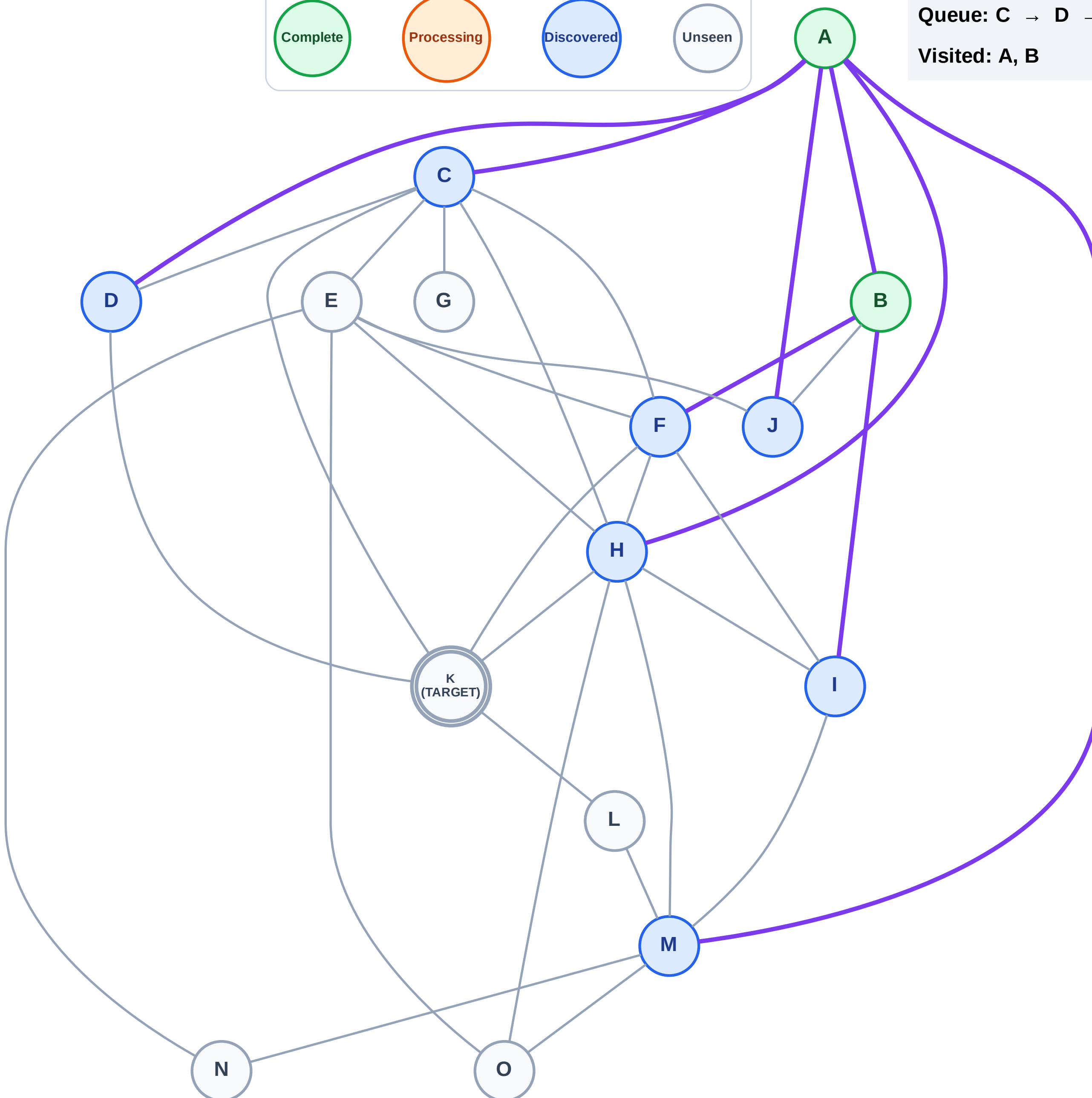
Discovered



Unseen

Queue: C → D → H → J → M → F → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

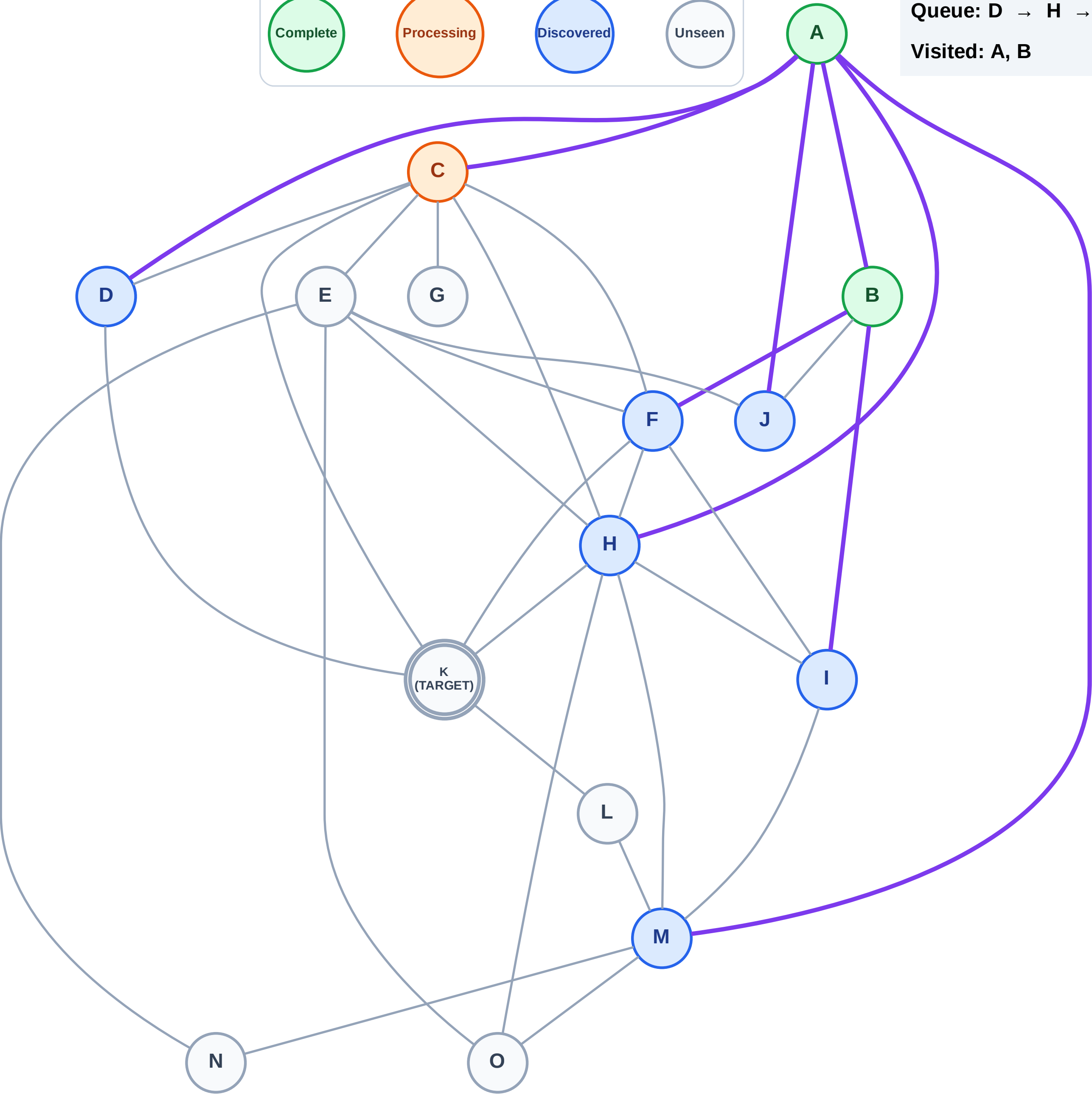
Processing

Discovered

Unseen

Queue: D → H → J → M → F → I

Visited: A, B



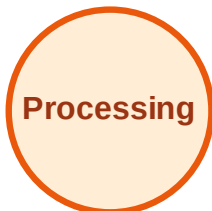
BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



Complete



Processing



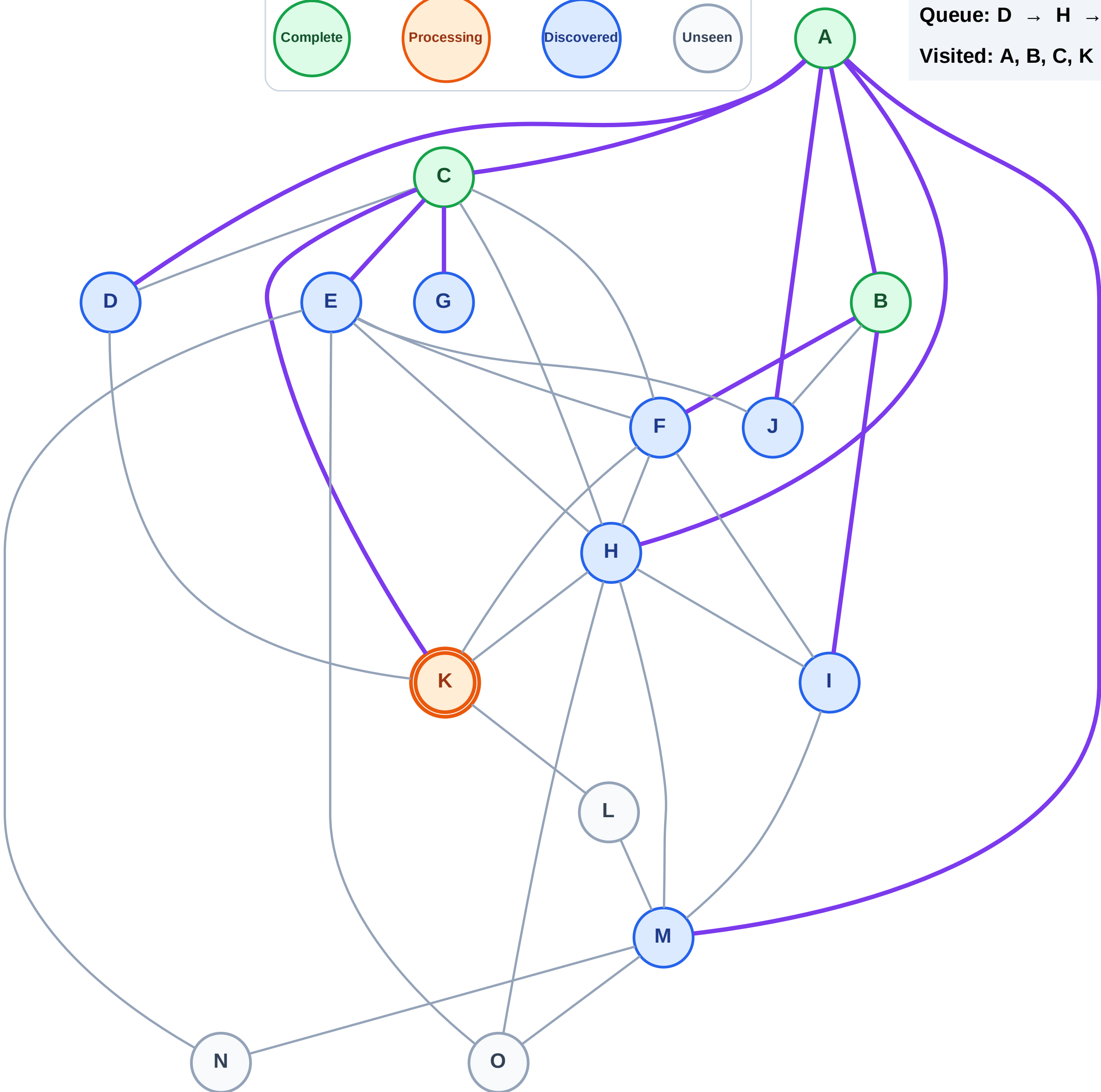
Discovered



Unseen

Queue: D → H → J → M → F → I → E → G

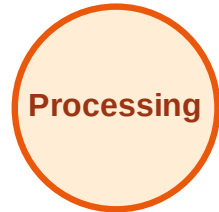
Visited: A, B, C, K



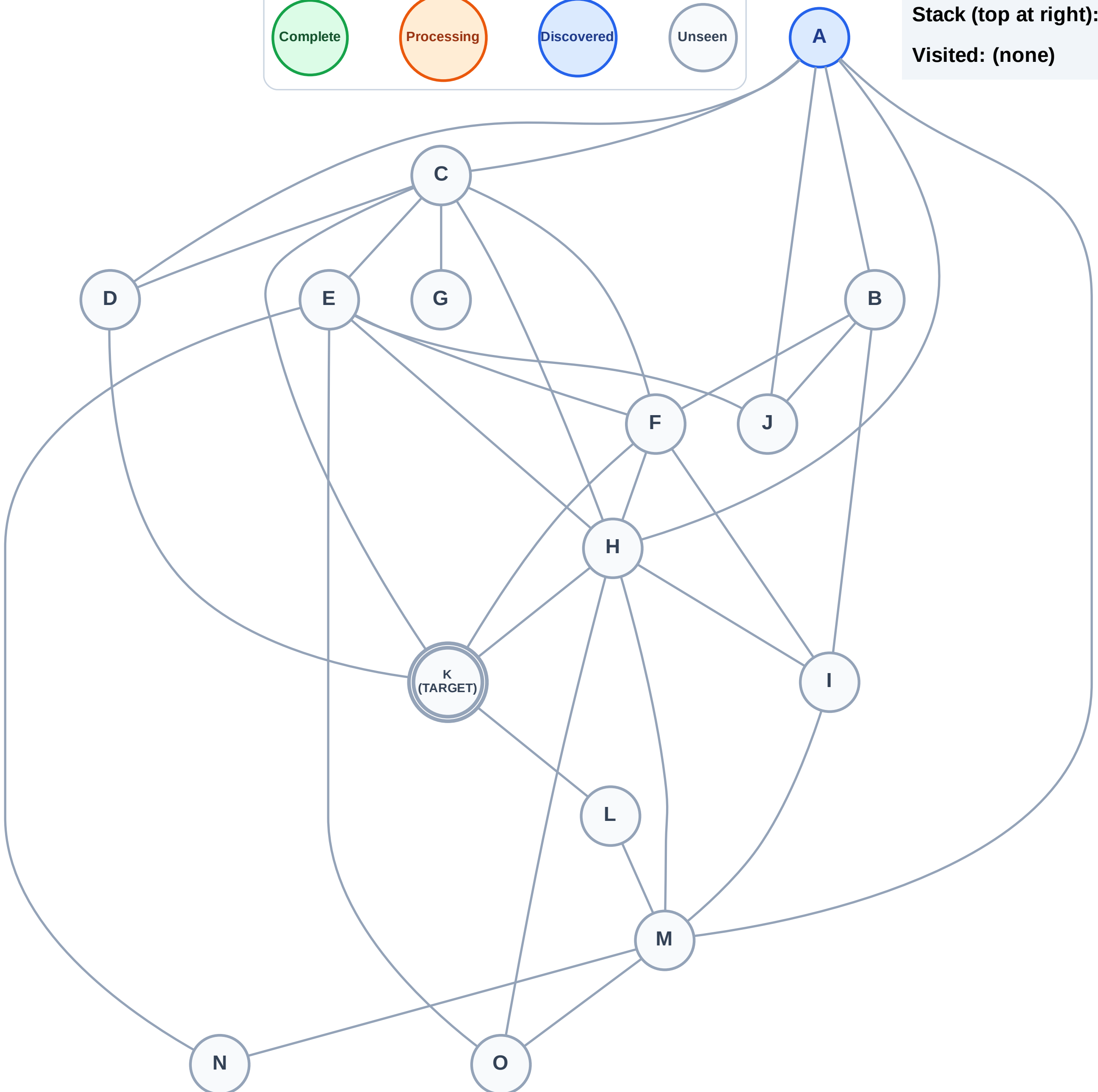
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



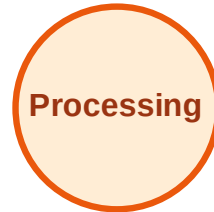
Stack (top at right): A
Visited: (none)



DFS Traversal

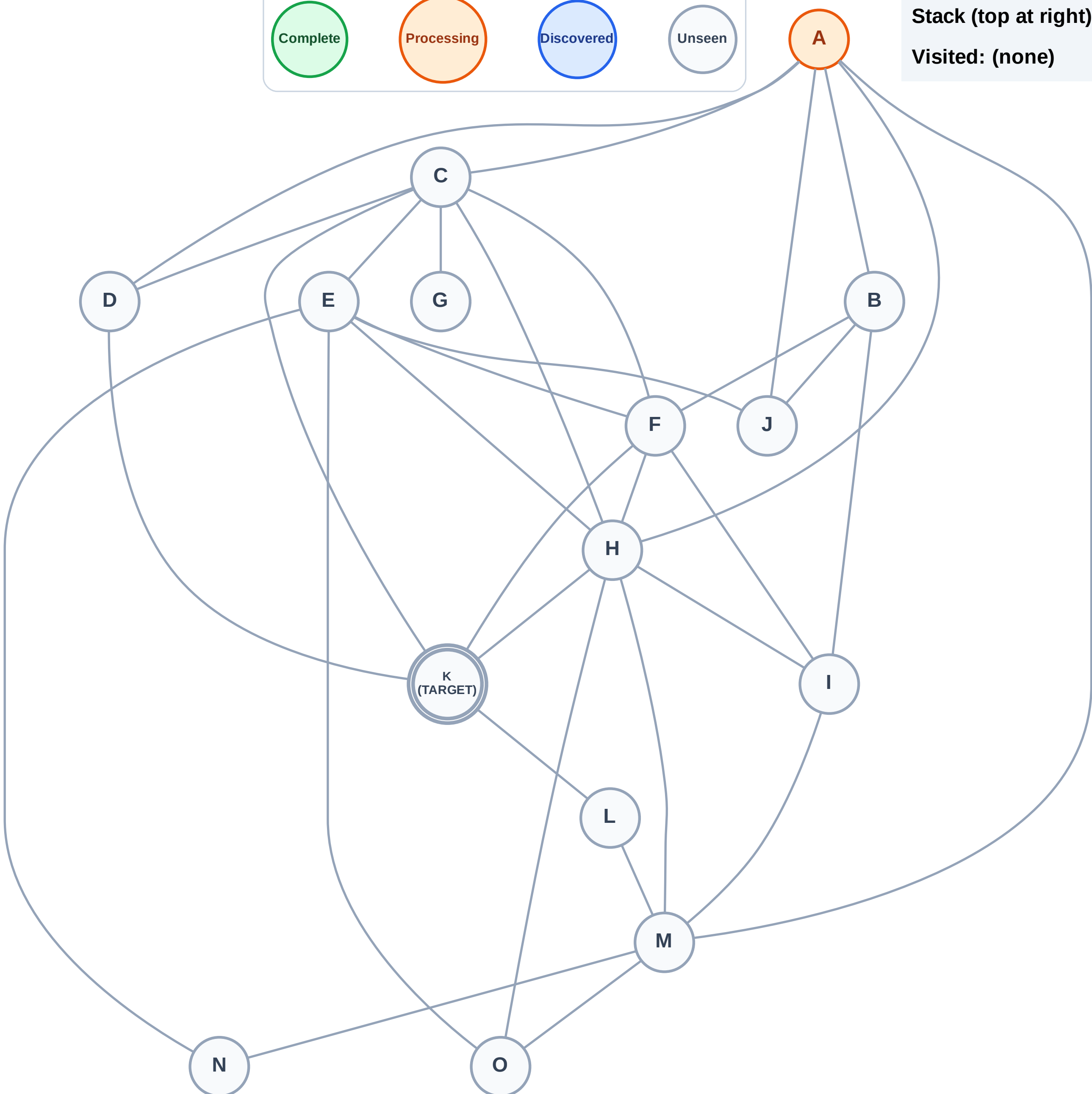
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover M, J, H, D, C, B from A

Legend

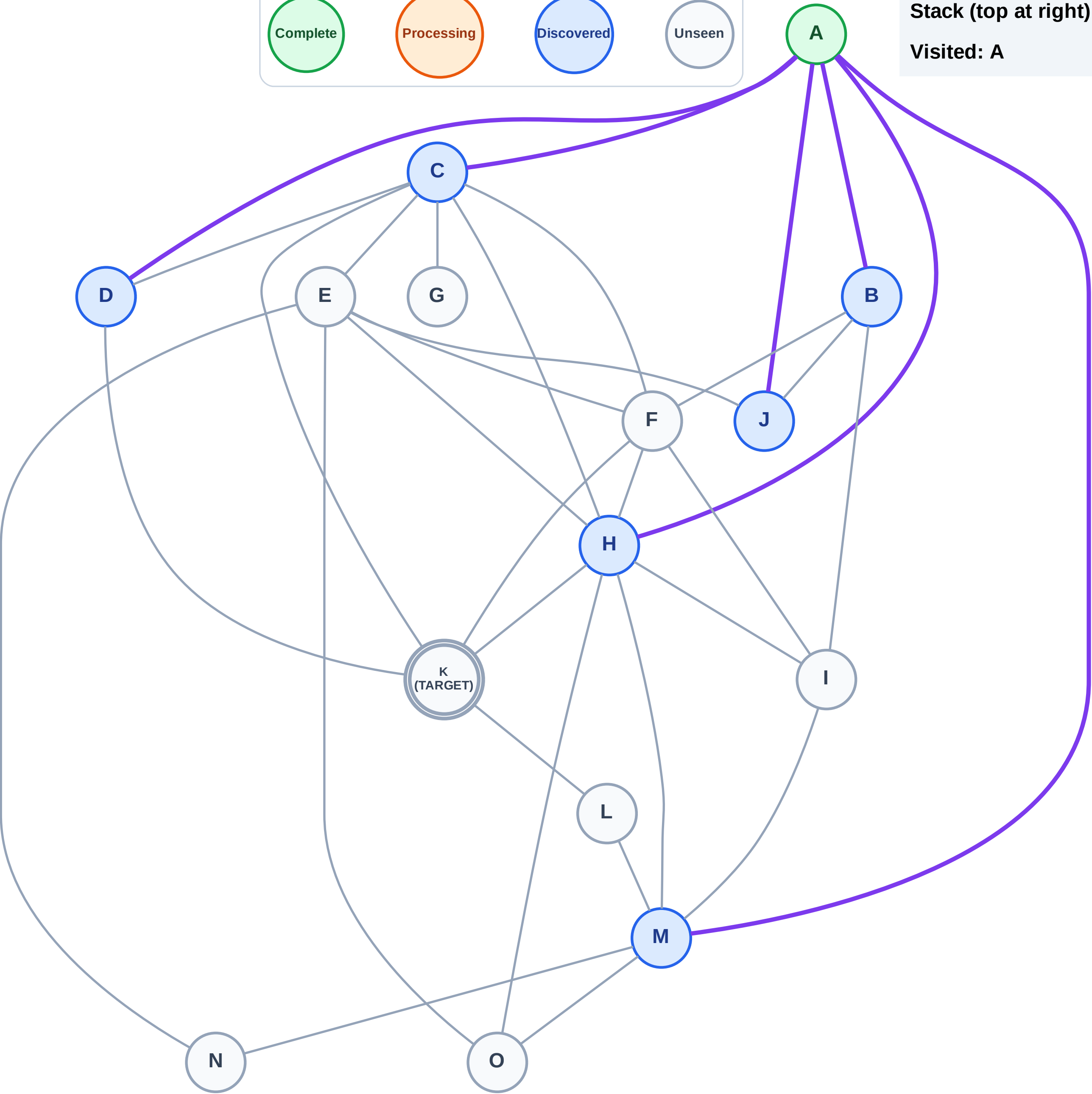
Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | D | C | B
Visited: A



Step 4: Process node B

Step 4: Process node B

Legend

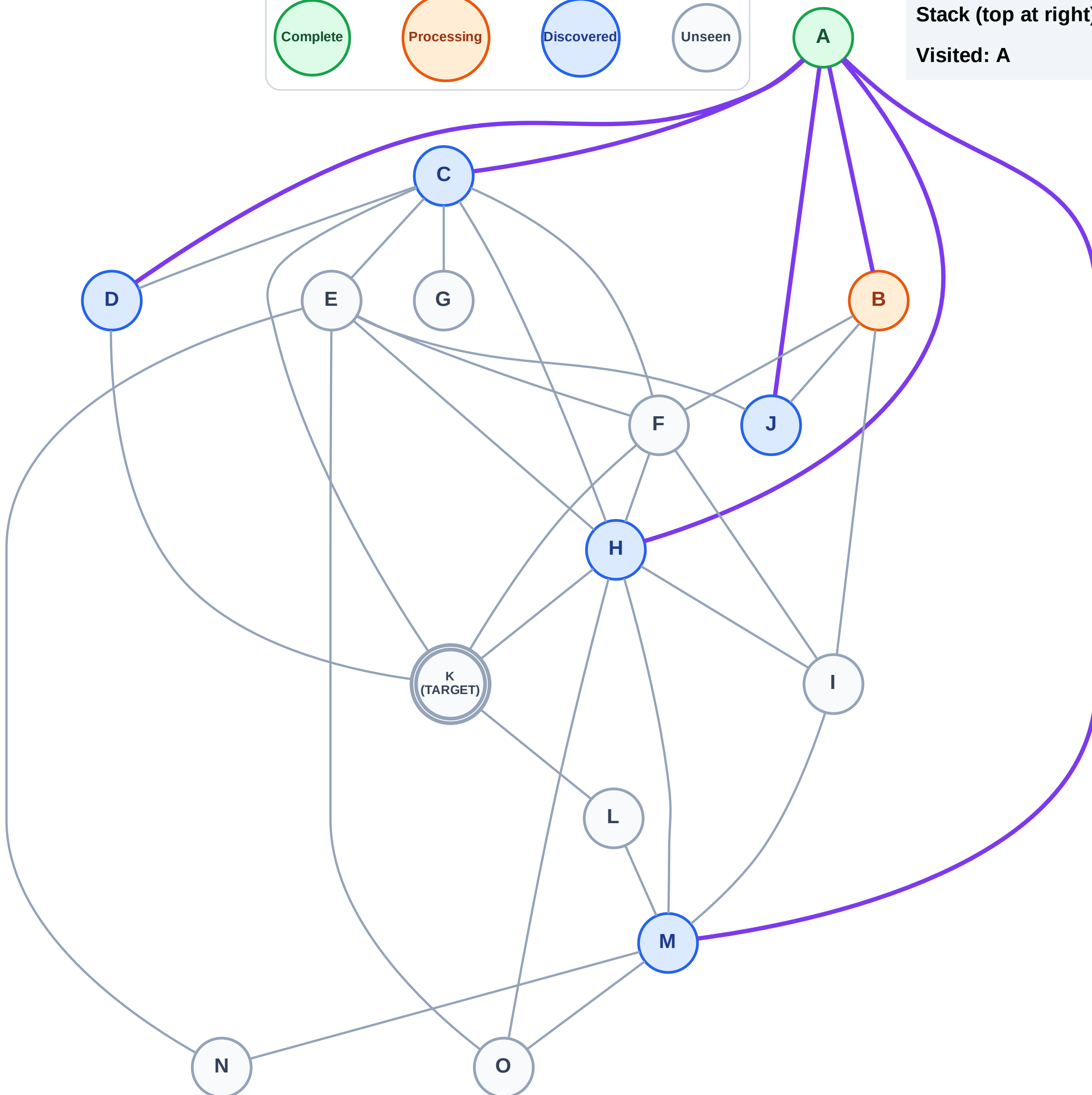


The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Processing

Unseen

Visited: A



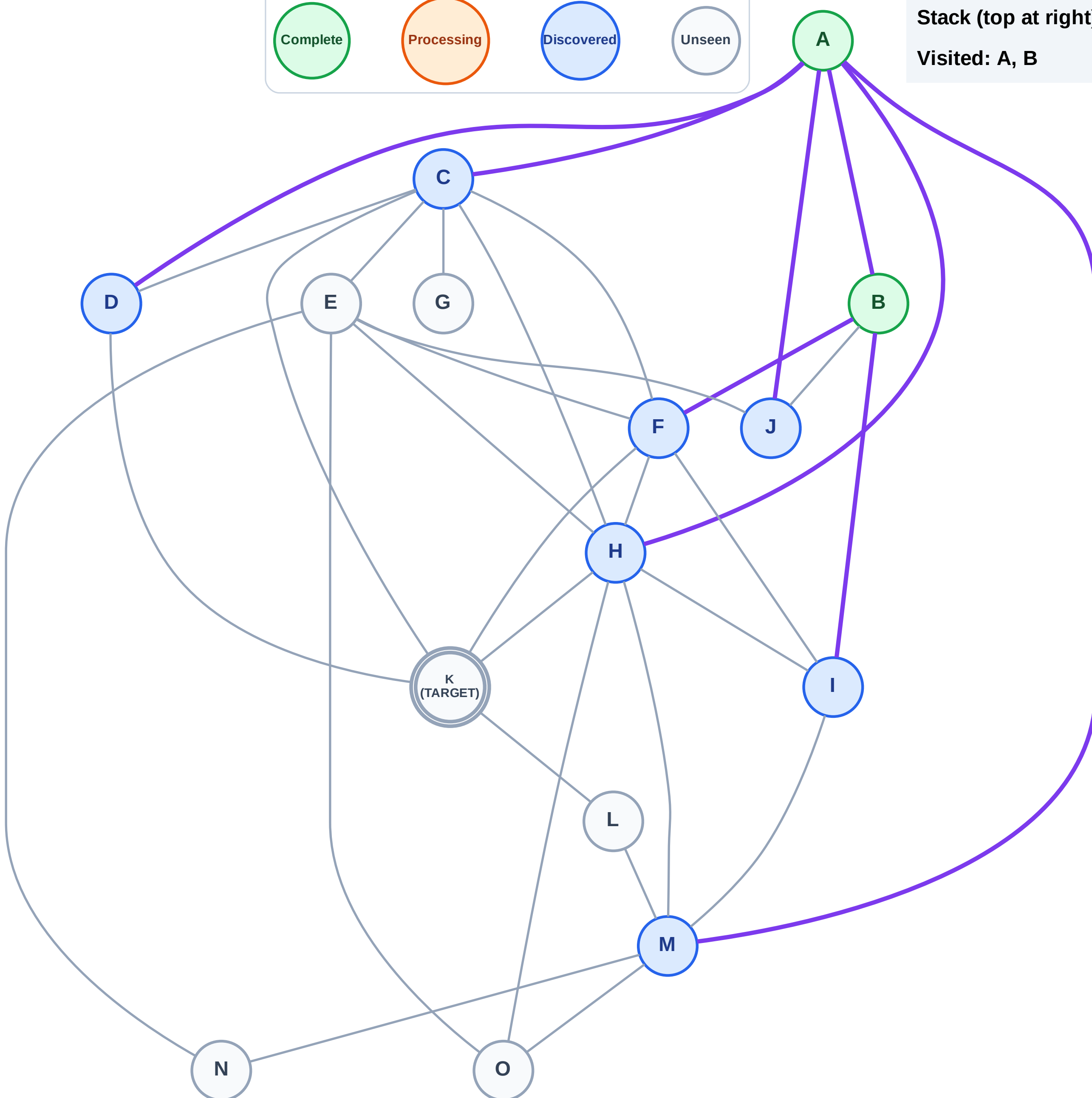
DFS Traversal

Step 5: Discover I, F from B

Legend



Stack (top at right): M | J | H | D | C | I | F
Visited: A, B



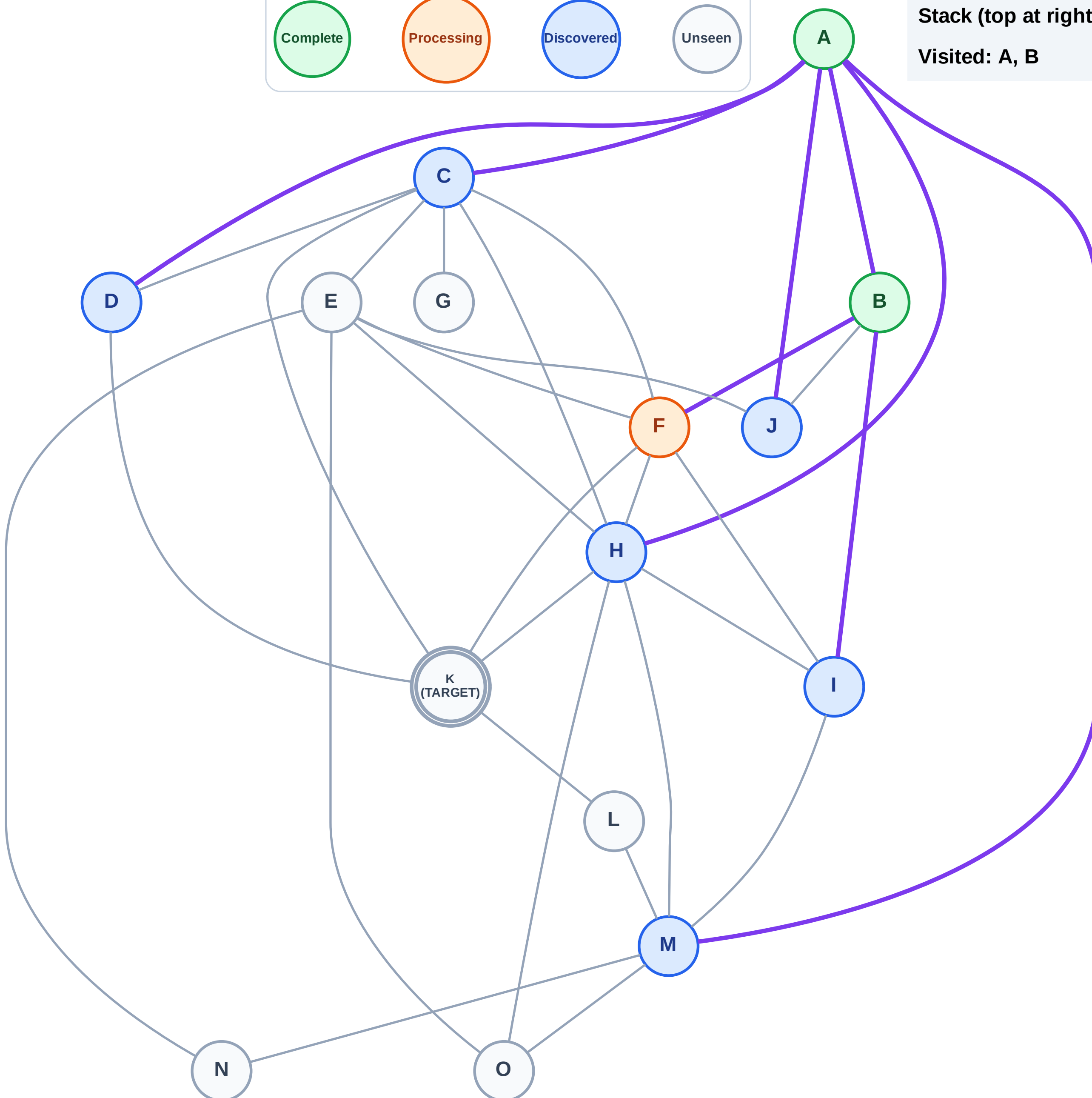
Step 6: Process node F

Step 6: Process node F

Complete Processing Discovered Unseen

Unseen

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | D | C | I

Visited: A, B, F, K

